

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Ordinary Level

MATHEMATICS 4004/1 PAPER 1 - November 2023

O-Level | Time: 2 hours 30 minutes | 30 Questions | 100 Marks | NON-CALCULATOR - Neither tables nor calculators may be used

Centre Number _____ Candidate Number _____ Name _____

INSTRUCTIONS: Answer ALL questions. Each question [1-3 marks]. Show all working. Additional materials: Mathematical instruments, Graph paper.

1. Evaluate $3 + 4 \times 2 - 5$ [2] (Directed numbers)

2. Express 0.0045 in standard form [1] (Scientific notation)

3. List the elements of set $A = \{x: x \text{ is prime } < 10\}$ [1] (Sets)

4. Simplify $3x + 2x - x$ [1] (Algebra)

5. Solve $2x + 3 = 11$ [2]

6. Factorise $x^2 - 5x + 6$ [2]

7. Find the HCF of 12 and 18 [1]

8. Convert 13 to binary [2]

9. Find the area of triangle base 8cm height 5cm [2]

10. Solve simultaneous $2x + y = 7$, $x - y = 2$ [3]

11. Find the gradient of line joining (2,3) and (4,7) [2]

12. Solve $3(x-2)=15$ [2]

13. Find the mean of 2,4,6,8,10 [2]

14. Simplify $(2a)^3$ [1]

15. Find interior angle of regular hexagon [2]

16. Solve inequality $2x - 3 < 7$ [2]

17. Volume of cylinder radius 7 height 10 ($\pi=22/7$) [3]

18. Factorise $4x^2 - 9$ [2]

19. Probability bag 3 red 2 blue [2]

20. Rationalise $2/\sqrt{3}$ [2]

21. Solve $x^2 - 3x - 4 = 0$ [3]

22. nth term sequence 3,7,11... [2]

23. Simple interest \$500 at 10% for 2 years [2]

24. Area of circle radius 7 [2]

25. Solve $\log_2 x = 3$ [1]

26. Median of 12,5,9,15,7 [2]

27. Expand $(x+3)(x-2)$ [2]

28. Time: 60km at 30km/h [2]

29. Express $5/8$ as decimal [1]

30. Find the LCM of 6 and 8 [1]